

EDUCATION

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| Mar 2021 | <b>Doctor of Philosophy (PhD)</b><br><i>Oregon Health &amp; Science University, Portland, OR</i><br><i>Field:</i> Bioinformatics and Computational Biomedicine<br><i>Dissertation Title:</i> Machine Learning for Disease Detection and Prediction in Retinopathy of Prematurity<br><i>Dissertation Advisory Committee:</i> Michael F. Chiang, MD, MA; J. Peter Campbell, MD, MPH; Kemal Sönmez, PhD; Jayashree Kalpathy-Cramer, PhD; Ted Laderas, PhD |
| Jun 2014 | <b>Bachelor of Science (BS)</b><br><i>Pacific Lutheran University, Tacoma, WA</i><br><i>Field:</i> Biochemistry                                                                                                                                                                                                                                                                                                                                        |

RESEARCH EXPERIENCE

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| Apr 2021–Present  | <b>Computational Biologist IV   2024–Present</b><br><b>Computational Biologist III   2021–2024</b><br><i>Oregon Health &amp; Science University, Portland, OR</i><br>Specializations: Machine Learning, Deep Learning, Imaging, Diagnostic and Predictive Modeling, Statistics, Python, R                            |
| May 2023–Present  | <b>Consultant</b><br><i>Siloam Vision</i><br>Specializations: Data Analysis, Statistics, and Predictive Modeling                                                                                                                                                                                                     |
| Jun 2016–Mar 2021 | <b>PhD Candidate</b><br><i>Oregon Health &amp; Science University, Portland, OR</i><br><i>Dissertation Title:</i> Machine Learning for Disease Detection and Prediction in Retinopathy of Prematurity<br><i>Specializations:</i> Deep Learning, Image Analysis, Retinopathy of Prematurity, Risk Modeling, Python, R |
| Aug 2014–Dec 2016 | <b>Research Assistant I–II</b><br><i>Oregon Health &amp; Science University, Portland, OR</i><br><i>Specializations:</i> Rodent Handling, Optical Coherence Tomography, Electroretinogram, Western Blot, Immunohistochemistry, Python, MATLAB                                                                        |
| Sep 2013–May 2014 | <b>Student Research Assistant</b><br><i>Pacific Lutheran University, Tacoma, WA</i><br><i>Specializations:</i> Polymer Synthesis, Chromatography, Spectroscopy, IGOR Pro, MATLAB                                                                                                                                     |

ADVISORY ROLES

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| May 2023–Present | <b>Scientific Advisory Board Member</b><br><i>Siloam Vision</i>                                                                                                                                        |
| Jan–Mar 2024     | <b>Data Science Consultant</b><br><i>Office of Data Science and Health Informatics, National Eye Institute</i><br>Provided insights into current data science methods and tools used in ophthalmology. |

HONORS AND AWARDS

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| <b>World Retinopathy of Prematurity Congress</b> |                                                                                                                                                                                                                                                                                    |
| Sep 2022                                         | Paper of the Year:<br><b>Coyner AS</b> , Chen JS, Singh P, et al. Single-Examination Risk Prediction of Severe Retinopathy of Prematurity. <i>Pediatrics</i> . 2021;148(6):e2021051772.                                                                                            |
| <b>American Medical Informatics Association</b>  |                                                                                                                                                                                                                                                                                    |
| Nov 2020                                         | Student Paper Competition Finalist                                                                                                                                                                                                                                                 |
| <b>Ophthalmology Retina</b>                      |                                                                                                                                                                                                                                                                                    |
| Dec 2019                                         | Top 3 Most Downloaded Pediatric Ophthalmology Paper of the Year:<br><b>Coyner AS</b> , Swan R, Campbell JP, et al. Automated Fundus Image Quality Assessment in Retinopathy of Prematurity Using Deep Convolutional Neural Networks. <i>Ophthalmol Retina</i> . 2019;3(5):444-450. |

### **Association for Research in Vision and Ophthalmology**

May 2018                      Members-in-Training Outstanding Poster Award Finalist  
May 2017                      Travel Grant Award from the National Eye Institute

### **Oregon Health & Science University**

Jun 2016–Mar 2021           National Library of Medicine Training Grant: T15 LM007088

### **Pacific Lutheran University**

Aug 2012–Jun 2014           Provost Merit Scholarship  
Aug 2012–Jun 2014           Academic Merit Award  
Aug 2012–Jun 2014           Phi Theta Kappa Scholarship  
Aug 2012–Jun 2014           Tacoma Rainiers Scholarship  
Aug 2012–Jun 2014           Larson, Ludwig, and Clara Scholarship

## **PROFESSIONAL AFFILIATIONS**

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2021–2022                      American Academy of Ophthalmology  
2021–2024                      American Association for Pediatric Ophthalmology and Strabismus  
2016–2022                      American Medical Informatics Association  
2014–*Present*                   Association for Research in Vision and Ophthalmology  
2012–2015                      American Chemical Society

## **REVIEWER EXPERIENCE**

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2020, 2022, 2026              Ophthalmology  
2026                              Ophthalmology Retina  
2025–2026                      Ophthalmology Science  
2024                              Retina  
2020, 2023–2024              Journal of the American Medical Informatics Association  
2021–2023                      JAMA Ophthalmology  
2023                              Nature Communications  
2023                              Scientific Data  
2019                              IEEE Transactions on Medical Imaging  
2018–2019                      Computers in Biology and Medicine

## **TEACHING EXPERIENCE**

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Oct 2022–Jun 2023           **Instructor**  
*Posit, PBC (formerly RStudio)*  
R and Python for Data Scientists

Nov 2020                      **Instructor**  
*American Academy of Ophthalmology*  
Artificial Intelligence: Demystification and Applications

Apr 2020–Jun 2020           **Teaching Assistant (Volunteer)**  
*Oregon Health & Science University, Portland, OR*  
Ready for R: An Introduction to Useful Data Workflows

Sep 2019–Dec 2019           **Instructor (Mentored Teaching)**  
*Oregon Health & Science University, Portland, OR*  
BMI 665: Bioinformatics Programming and Scripting

Jun 2019–Aug 2019           **Instructor (Mentored Teaching)**  
*Oregon Health & Science University, Portland, OR*  
BMI 669: Data Analytics

Apr 2019–Jun 2019           **Teaching Assistant**  
*Oregon Health & Science University, Portland, OR*  
CS 659: Machine Learning

Sep 2018–Dec 2018           **Teaching Assistant**  
*Oregon Health & Science University, Portland, OR*  
MATH 630: Probability & Statistical Inference for Scientists and Engineers

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|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| Jun 2018–Sep 2018 | <b>Teaching Assistant</b><br>Oregon Health & Science University, Portland, OR<br>BSTA 525: Introduction to Biostatistics |
| Nov 2011–Dec 2012 | <b>Peer Tutor</b><br>Pierce College, Lakewood, WA<br>Calculus, Physics, General Chemistry, and Organic Chemistry         |

## VOLUNTEER EXPERIENCE

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2015–Sep 2017 | <b>Casey Outreach Van</b><br>Provided vision screening services for under-served communities in Portland, Gresham, and Hillsboro.          |
| Jul 2013–Aug 2013 | <b>VIDA Volunteer Services</b><br>Provided medical services and patient education to under-served communities in Costa Rica and Nicaragua. |
| Sep 2011–Dec 2013 | <b>Guide Dogs for the Blind</b><br>Raised and trained two guide dogs for the visually impaired community.                                  |

## PATENT APPLICATIONS

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- Chiang MF, Campbell JP, Ostmo S, Chan RVP, Kalpathy-Cramer J, Brown JM, Erdogmus D, Ioannidis S, **Coyner AS**. Systems, Devices, and Methods for Identifying Plus Disease Using Deep Convolutional Neural Networks. U.S. Provisional Patent Application No. 62/814,188, filed March 5, 2019. Licensed to Siloam Vision.

## PEER-REVIEWED PUBLICATIONS

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- Campbell JP, Nayak S, Ostmo S, **Coyner AS**, Singh P, Kalpathy-Cramer J, Shah P, Chan RP, Chiang MF. Application of a Quantitative Vascular Severity Score in Retinopathy of Prematurity in the United States and India: New Insights Into Disease Epidemiology and Pathophysiology. *Am J Ophthalmol*. 2026 Oct;290:123-136. doi: 10.1016/j.ajo.2026.06.008. PMID: 42288327; PMCID: PMC13377564.
- Luong HN, **Coyner AS**, Roti EV, Sutter DA, Ostmo S, Jackson J, Woodward M, Bayhaqi Y, Ni S, Jian Y, Campbell JP, Young BK. Evaluation of Hyaloid Artery Persistence as a Biomarker in Retinopathy of Prematurity Using OCT. *Ophthalmol Sci*. 2026 Aug;6(8):101282. doi: 10.1016/j.xops.2026.101282. PMID: 42471851; PMCID: PMC13380042.
- Sutter DA, **Coyner AS**, Jackson J, Ni S, Woodward MK, Bayhaqi YA, Ostmo S, Scruggs BA, Hubbard GB, Dedania VS, Grigorian F, Miller AM, Eton E, Rishi P, Bhatt A, Cole E, Chan RVP, Chiang M, Jian Y, Campbell JP, Young BK. Nasal-Temporal Asymmetry in Retinopathy of Prematurity as a Source of Diagnostic Bias and Error. *Ophthalmol Retina*. 2026 Aug;10(8):892-895. doi: 10.1016/j.oret.2026.04.013. PMID: 42061569.
- Placide J, **Coyner AS**, Ostmo SR, Moshfeghi DM, Chiang MF, Kim SJ, Kalpathy-Cramer J, Chan RVP, Young BK, Campbell JP. Precision Risk Model Using Quantitative Assessment of Vascular Severity in Telemedicine-Based Screening. *JAMA Ophthalmol*. 2026 May;144(5):424-430. doi: 10.1001/jamaophthalmol.2026.0510. PMID: 41926091; PMCID: PMC13047491.
- Austeng D, Moljord K, Morken TS, **Coyner AS**, Campbell JP. Inter-observer variability is unlikely to explain regional differences in retinopathy of prematurity in Norway. *Acta Ophthalmol*. 2026 Apr 24. doi: 10.1111/aos.70146. PMID: 42032862.
- Hoebel KV, Lemay A, Campbell JP, Ostmo S, Chiang MF, Bridge CP, Li MD, Singh P, **Coyner AS**, Kalpathy-Cramer J. Leveraging deep learning to infer continuous predictions from ordinal labels in medical imaging. *PLOS Digit Health*. 2026 Apr;5(4):e0001248. doi: 10.1371/journal.pdig.0001248. PMID: 41996486; PMCID: PMC13089899.
- Nayak S, **Coyner AS**, Ostmo SR, Sutter D, Jackson JD, Bayhaqi Y, Jang JH, Kalpathy-Cramer J, Chan RP, Chiang MF, Young BK, Campbell JP. Quantitative Assessment of the Extent of Retinal Vascularization in Retinopathy of Prematurity. *Ophthalmol Sci*. 2026 Apr;6(4):101114. doi: 10.1016/j.xops.2026.101114. PMID: 41847074; PMCID: PMC12990333.
- Jackson JD, Woodward MK, Sutter DA, **Coyner AS**, Wang CX, Ostmo SR, Chiang MF, Jia Y, Huang D, Jian Y, Campbell JP, Young BK. Automated Feature Segmentation of Ultra-Widefield OCT Images. *Ophthalmol Sci*. 2026 Mar;6(3):101029. doi: 10.1016/j.xops.2025.101029. PMID: 41625354; PMCID: PMC12858344.
- Singh P, Kumar S, Tyagi R, Young BK, Jordan BK, Scottoline B, Evers PD, Ostmo S, **Coyner AS**, Lin WC, Gupta A, Erdogmus D, Chan RVP, McCourt EA, Barry JS, McEvoy CT, Chiang MF, Campbell JP, Kalpathy-Cramer J. Deep Learning-Based Prediction of Cardiopulmonary Disease in Retinal Images of Premature Infants. *JAMA Ophthalmol*. 2026 Mar;144(3):230-237. doi: 10.1001/jamaophthalmol.2025.5814. PMID: 41569552; PMCID: PMC12828656.

10. Sutter DA, Igelman A, **Coyner AS**, Stratis G, Real K, Woodward MK, Jackson J, Bayhaqi Y, Ni S, Ostmo S, Jian Y, Chiang MF, Campbell JP, Guidoboni G, Young BK. Computational Vascular Development Model Explaining Incidence of Notch in Retinopathy of Prematurity With Ultra-Widefield Optical Coherence Tomography. *Invest Ophthalmol Vis Sci*. 2026 Feb;67(2):10. doi: 10.1167/iovs.67.2.10. PMID: 41631756; PMCID: PMC12882098.
11. Sutter DA, Woodward MK, Jackson J, Bayhaqi Y, **Coyner AS**, Ni S, Ostmo S, Lima TT, Nagiel A, Chiang MF, Young BK, Jian Y, Campbell JP. OCT-Derived Quantitative Measurement of Extent of Vascularization ("Zone") in Retinopathy of Prematurity. *Ophthalmol Sci*. 2026 Jan;6(1):100912. doi: 10.1016/j.xops.2025.100912. PMID: 41140894; PMCID: PMC12548099.
12. Sutter DA, Woodward MK, Jackson J, Bayhaqi Y, **Coyner AS**, Ni S, Ostmo S, Chiang MF, Young BK, Jian Y, Campbell JP. Topographic Features in Retinopathy of Prematurity With en Face Ultra-Widefield OCT. *Ophthalmol Retina*. 2026 Jan;10(1):88-94. doi: 10.1016/j.oret.2025.07.003. PMID: 40639465; PMCID: PMC12381962.
13. Song A, Lee E, Ostmo SR, Jonas KE, Chervinko MA, **Coyner AS**, Campbell JP, Chan RVP. Retinopathy of prematurity training and education: A systematic review. *Surv Ophthalmol*. 2026;71(1):248-261. doi: 10.1016/j.survophthal.2025.06.007. PMID: 40541843; PMCID: PMC12718762.
14. Derks LA, Tekin YS, Loudon SE, Vingerling JR, **Coyner AS**, Campbell JP, Tjiam AM. External validation of an artificial intelligence-based model for retinopathy of prematurity screening using Phoenix ICON retinal images. *J AAPOS*. 2025 Dec;29(6):104696. doi: 10.1016/j.jaapos.2025.104696. PMID: 41276153; PMCID: PMC13345673.
15. Burt SS, **Coyner AS**, Roti EV, Bayhaqi Y, Jackson J, Woodward MK, Ni S, Ostmo SR, Liang G, Jia Y, Huang D, Chiang MF, Young BK, Jian Y, Campbell JP. Automated Quantification of Retinopathy of Prematurity Stage via Ultrawidefield OCT. *Ophthalmol Sci*. 2025;5(2):100663. doi: 10.1016/j.xops.2024.100663. PMID: 39866345; PMCID: PMC11760822.
16. Roti EV, Ni S, Bayhaqi Y, Ostmo SR, Burt SS, Woodward MK, Jackson JD, **Coyner AS**, Chiang MF, Huang D, Campbell JP, Jian Y, Young BK. Ultra-Widefield Optical Coherence Tomography Beyond the Ora Serrata in Retinopathy of Prematurity. *JAMA Ophthalmol*. 2025 Feb;143(2):165-170. doi: 10.1001/jamaophthalmol.2024.5533. PMID: 39724290; PMCID: PMC11843357.
17. **Coyner AS**, Young BK, Ostmo SR, Grigorian F, Ells A, Hubbard B, Rodriguez SH, Rishi P, Miller AM, Bhatt AR, Agarwal-Sinha S, Sears J, Chan RVP, Chiang MF, Kalpathy-Cramer J, Binenbaum G, Campbell JP. Use of an Artificial Intelligence-Generated Vascular Severity Score Improved Plus Disease Diagnosis in Retinopathy of Prematurity. *Ophthalmology*. 2024 Nov;131(11):1290-1296. doi: 10.1016/j.ophtha.2024.06.006. PMID: 38866367; PMCID: PMC11499038.
18. Binenbaum G, Stahl A, **Coyner AS**, He J, Ying GS, Ostmo S, Chan RVP, Toth C, Vinekar A, Campbell JP, International Classification of Retinopathy of Prematurity Committee. P Score: A Reference Image-Based Clinical Grading Scale for Vascular Change in Retinopathy of Prematurity. *Ophthalmology*. 2024 Nov;131(11):1297-1303. doi: 10.1016/j.ophtha.2024.05.019. PMID: 38795976; PMCID: PMC11499040.
19. Burt SS, Woodward M, Ni S, Jackson J, **Coyner AS**, Ostmo SR, Liang G, Bayhaqi Y, Jia Y, Huang D, Chiang MF, Young BK, Jian Y, Campbell JP. Isolated Retinal Neovascularization in Retinopathy of Prematurity: Clinical Associations and Prognostic Implications. *Ophthalmol Retina*. 2024 Oct;8(10):1021-1028. doi: 10.1016/j.oret.2024.04.025. PMID: 38735640; PMCID: PMC11446652.
20. Lin WC, **Coyner AS**, Amankwa CE, Lucero A, Wollstein G, Schuman JS, Ishikawa H. High Prevalence of Artifacts in Optical Coherence Tomography With Adequate Signal Strength. *Transl Vis Sci Technol*. 2024 Aug;13(8):43. doi: 10.1167/tvst.13.8.43. PMID: 39196579; PMCID: PMC11364177.
21. **Coyner AS**, Murickan T, Oh MA, Young BK, Ostmo SR, Singh P, Chan RVP, Moshfeghi DM, Shah PK, Venkatapathy N, Chiang MF, Kalpathy-Cramer J, Campbell JP. Multinational External Validation of Autonomous Retinopathy of Prematurity Screening. *JAMA Ophthalmol*. 2024 Apr;142(4):327-335. doi: 10.1001/jamaophthalmol.2024.0045. PMID: 38451496; PMCID: PMC10921347.
22. Lin WC, Jordan BK, Scottoline B, Ostmo SR, **Coyner AS**, Singh P, Kalpathy-Cramer J, Erdogmus D, Chan RVP, Chiang MF, Campbell JP. Oxygenation Fluctuations Associated with Severe Retinopathy of Prematurity: Insights from a Multimodal Deep Learning Approach. *Ophthalmol Sci*. 2024;4(2):100417. doi: 10.1016/j.xops.2023.100417. PMID: 38059124; PMCID: PMC10696464.
23. de Oliveira TB, Young BK, Ostmo SR, **Coyner AS**, Gensure RH, Scruggs BA, Anderson J, Moshfeghi D, Campbell JP. Macular Thickness Is Associated with Persistent Avascular Retina in Patients with a History of Untreated Retinopathy of Prematurity. *Ophthalmol Retina*. 2024 Feb;8(2):202-204. doi: 10.1016/j.oret.2023.10.004. PMID: 37838277; PMCID: PMC10873111.
24. Li X, Owen LA, Taylor KD, Ostmo S, Chen YI, **Coyner AS**, Sonmez K, Hartnett ME, Guo X, Ipp E, Roll K, Genter P, Chan RVP, DeAngelis MM, Chiang MF, Campbell JP, Rotter JI, i-ROP Consortium. Genome-wide association identifies novel ROP risk loci in a multiethnic cohort. *Commun Biol*. 2024 Jan;7(1):107. doi: 10.1038/s42003-023-05743-9. PMID: 38233474; PMCID: PMC10794688.

25. Chen JS, Marra KV, Robles-Holmes HK, Ly KB, Miller J, Wei G, Aguilar E, Bucher F, Ideguchi Y, **Coyner AS**, Ferrara N, Campbell JP, Friedlander M, Nudleman E. Applications of Deep Learning: Automated Assessment of Vascular Tortuosity in Mouse Models of Oxygen-Induced Retinopathy. *Ophthalmol Sci.* 2024;4(1):100338. doi: 10.1016/j.xops.2023.100338. PMID: 37869029; PMCID: PMC10585474.
26. Robles-Holmes H, **Coyner AS**, Campbell JP, Nudleman E. Imaging Features Associated with Persistent Avascular Retina in Retinopathy of Prematurity. *Ophthalmology.* 2024 Jan;131(1):122-124. doi: 10.1016/j.ophtha.2023.09.032. PMID: 37805003.
27. deCampos-Stairiker MA\*, **Coyner AS\***, Gupta A, Oh M, Shah PK, Subramanian P, Venkatapathy N, Singh P, Kalpathy-Cramer J, Chiang MF, Chan RVP, Campbell JP. Epidemiologic Evaluation of Retinopathy of Prematurity Severity in a Large Telemedicine Program in India Using Artificial Intelligence. *Ophthalmology.* 2023 Aug;130(8):837-843. doi: 10.1016/j.ophtha.2023.03.026. PMID: 37030453; PMCID: PMC10524227. \*Equal contribution.
28. Young BK, Cole ED, Shah PK, Ostmo S, Subramaniam P, Venkatapathy N, Tsai ASH, **Coyner AS**, Gupta A, Singh P, Chiang MF, Kalpathy-Cramer J, Chan RVP, Campbell JP. Efficacy of Smartphone-Based Telescreening for Retinopathy of Prematurity With and Without Artificial Intelligence in India. *JAMA Ophthalmol.* 2023 Jun;141(6):582-588. doi: 10.1001/jamaophthalmol.2023.1466. PMID: 37166816; PMCID: PMC10176185.
29. **Coyner AS**, Singh P, Brown JM, Ostmo S, Chan RVP, Chiang MF, Kalpathy-Cramer J, Campbell JP; Imaging and Informatics in Retinopathy of Prematurity Consortium. Association of Biomarker-Based Artificial Intelligence With Risk of Racial Bias in Retinal Images. *JAMA Ophthalmol.* 2023 Jun;141(6):543-552. doi: 10.1001/jamaophthalmol.2023.1310. PMID: 37140902; PMCID: PMC10160994.
30. Eilts SK, Pfeil JM, Poschkamp B, Krohne TU, Eter N, Barth T, Guthoff R, Lagrèze W, Grundel M, Bründer MC, Busch M, Kalpathy-Cramer J, Chiang MF, Chan RVP, **Coyner AS**, Ostmo S, Campbell JP, Stahl A; Comparing Alternative Ranibizumab Dosages for Safety and Efficacy in Retinopathy of Prematurity (CARE-ROP) Study Group. Assessment of Retinopathy of Prematurity Regression and Reactivation Using an Artificial Intelligence-Based Vascular Severity Score. *JAMA Netw Open.* 2023 Jan;6(1):e2251512. doi: 10.1001/jamanetworkopen.2022.51512. PMID: 36656578; PMCID: PMC9857423.
31. Cole E, Valikodath NG, Al-Khaled T, Bajimaya S, Kc S, Chuluunbat T, Munkhuu B, Jonas KE, Chuluunkhuu C, MacKeen LD, Yap V, Hallak J, Ostmo S, Wu WC, **Coyner AS**, Singh P, Kalpathy-Cramer J, Chiang MF, Campbell JP, Chan RVP. Evaluation of an Artificial Intelligence System for Retinopathy of Prematurity Screening in Nepal and Mongolia. *Ophthalmol Sci.* 2022 Dec;2(4):100165. doi: 10.1016/j.xops.2022.100165. PMID: 36531583; PMCID: PMC9754980.
32. Nguyen TP, Ni S, Ostmo S, Rajagopalan A, **Coyner AS**, Woodward M, Chiang MF, Jia Y, Huang D, Campbell JP, Jian Y. Association of Optical Coherence Tomography-Measured Fibrovascular Ridge Thickness and Clinical Disease Stage in Retinopathy of Prematurity. *JAMA Ophthalmol.* 2022 Nov;140(11):1121-1127. doi: 10.1001/jamaophthalmol.2022.4173. PMID: 36227622; PMCID: PMC9562098.
33. **Coyner AS**, Oh MA, Shah PK, Singh P, Ostmo S, Valikodath NG, Cole E, Al-Khaled T, Bajimaya S, K C S, Chuluunbat T, Munkhuu B, Subramanian P, Venkatapathy N, Jonas KE, Hallak JA, Chan RVP, Chiang MF, Kalpathy-Cramer J, Campbell JP. External Validation of a Retinopathy of Prematurity Screening Model Using Artificial Intelligence in 3 Low- and Middle-Income Populations. *JAMA Ophthalmol.* 2022 Aug;140(8):791-798. doi: 10.1001/jamaophthalmol.2022.2135. PMID: 35797036; PMCID: PMC9264225.
34. Hanif A, Lu C, Chang K, Singh P, **Coyner AS**, Brown JM, Ostmo S, Chan RVP, Rubin D, Chiang MF, Kalpathy-Cramer J, Campbell JP; Imaging and Informatics in Retinopathy of Prematurity Consortium. Federated Learning for Multicenter Collaboration in Ophthalmology: Implications for Clinical Diagnosis and Disease Epidemiology. *Ophthalmol Retina.* 2022 Aug;6(8):650-656. doi: 10.1016/j.oret.2022.03.005. PMID: 35304305; PMCID: PMC9357070.
35. Lu C, Hanif A, Singh P, Chang K, **Coyner AS**, Brown JM, Ostmo S, Chan RVP, Rubin D, Chiang MF, Campbell JP, Kalpathy-Cramer J; Imaging and Informatics in Retinopathy of Prematurity Consortium. Federated Learning for Multicenter Collaboration in Ophthalmology: Improving Classification Performance in Retinopathy of Prematurity. *Ophthalmol Retina.* 2022 Aug;6(8):657-663. doi: 10.1016/j.oret.2022.02.015. PMID: 35296449.
36. **Coyner AS**, Chen JS, Chang K, Singh P, Ostmo S, Chan RVP, Chiang MF, Kalpathy-Cramer J, Campbell JP; Imaging and Informatics in Retinopathy of Prematurity Consortium. Synthetic Medical Images for Robust, Privacy-Preserving Training of Artificial Intelligence: Application to Retinopathy of Prematurity Diagnosis. *Ophthalmol Sci.* 2022 Jun;2(2):100126. doi: 10.1016/j.xops.2022.100126. PMID: 36249693; PMCID: PMC9560638.
37. Chen JS, Anderson JE, **Coyner AS**, Ostmo S, Sonmez K, Erdogmus D, Jordan BK, McEvoy CT, Dukhovny D, Schelonka RL, Chan RVP, Singh P, Kalpathy-Cramer J, Chiang MF, Campbell JP. Quantification of Early Neonatal Oxygen Exposure as a Risk Factor for Retinopathy of Prematurity Requiring Treatment. *Ophthalmol Sci.* 2021 Dec;1(4):100070. doi: 10.1016/j.xops.2021.100070. PMID: 36275192; PMCID: PMC9562374.
38. Chen JS, **Coyner AS**, Chan RVP, Hartnett ME, Moshfeghi DM, Owen LA, Kalpathy-Cramer J, Chiang MF, Campbell JP. Deepfakes in Ophthalmology: Applications and Realism of Synthetic Retinal Images from Generative Adversarial Networks. *Ophthalmol Sci.* 2021 Dec;1(4):100079. doi: 10.1016/j.xops.2021.100079. PMID: 36246951; PMCID: PMC9562356.

39. **Coyner AS**, Chen JS, Singh P, Schelonka RL, Jordan BK, McEvoy CT, Anderson JE, Chan RVP, Sonmez K, Erdogmus D, Chiang MF, Kalpathy-Cramer J, Campbell JP. Single-Examination Risk Prediction of Severe Retinopathy of Prematurity. *Pediatrics*. 2021 Dec;148(6):e2021051772. doi: 10.1542/peds.2021-051772. PMID: 34814160; PMCID: PMC8919718.
40. Chen JS, **Coyner AS**, Ostmo S, Sonmez K, Bajimaya S, Pradhan E, Valikodath N, Cole ED, Al-Khaled T, Chan RVP, Singh P, Kalpathy-Cramer J, Chiang MF, Campbell JP. Deep Learning for the Diagnosis of Stage in Retinopathy of Prematurity: Accuracy and Generalizability across Populations and Cameras. *Ophthalmol Retina*. 2021 Oct;5(10):1027-1035. doi: 10.1016/j.oret.2020.12.013. PMID: 33561545; PMCID: PMC8364291.
41. **Coyner AS**, Chen J, Campbell JP, Ostmo S, Singh P, Kalpathy-Cramer J, Chiang MF. Diagnosability of Synthetic Retinal Fundus Images for Plus Disease Detection in Retinopathy of Prematurity. *AMIA Annu Symp Proc*. 2021 Jan;2020:329-337. PMID: 33936405; PMCID: PMC8075515.
42. Choi RY\*, **Coyner AS\***, Kalpathy-Cramer J, Chiang MF, Campbell JP. Introduction to Machine Learning, Neural Networks, and Deep Learning. *Transl Vis Sci Technol*. 2020 Feb;9(2):14. doi: 10.1167/tvst.9.2.14. PMID: 32704420; PMCID: PMC7347027. \*Equal contribution.
43. **Coyner AS**, Swan R, Campbell JP, Ostmo S, Brown JM, Kalpathy-Cramer J, Kim SJ, Jonas KE, Chan RVP, Chiang MF; Imaging and Informatics in Retinopathy of Prematurity Research Consortium. Automated Fundus Image Quality Assessment in Retinopathy of Prematurity Using Deep Convolutional Neural Networks. *Ophthalmol Retina*. 2019 May;3(5):444-450. doi: 10.1016/j.oret.2019.01.015. PMID: 31044738; PMCID: PMC6501831.
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## CONFERENCE PRESENTATIONS

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1. **Coyner AS**, Jang JH, Young BK, *et al*. Early vascular and choroidal trajectories precede macular edema in retinopathy of prematurity. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2026. Denver, CO. Oral presentation.
2. **Coyner AS**, Bayhaqi Y, Young BK, *et al*. En face OCT super-resolution in retinopathy of prematurity: enhanced low-resolution images for faster screening. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2025. Salt Lake City, UT. Poster presentation.
3. **Coyner AS**, Young BK, Ostmo SR, *et al*. Assistive artificial intelligence for improved clinical diagnosis of retinopathy of prematurity. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2024. Seattle, WA. Oral presentation.
4. **Coyner AS**, Lemay A, Hoebel K, *et al*. Monte Carlo dropout for increased deep learning repeatability and disease classification performance in retinopathy of prematurity. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2023. New Orleans, LA. Oral presentation.
5. **Coyner AS**, Cole ED, Young BK, *et al*. Artificial intelligence-assisted detection of referral-warranted retinopathy of prematurity using smartphone-based retinal fundus cameras. *American Association for Pediatric Ophthalmology and Strabismus Annual Meeting*. March 2023. New York, NY. Poster presentation.

6. **Coyner AS**, Singh P, Brown JM, *et al.* Grayscale Retinal Vessel Maps Are Associated with Self-Reported Race: Implications for Artificial Intelligence Models. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2022. Denver, CO. Oral presentation.
7. **Coyner AS**, Oh MA, Shah PK, *et al.* Evaluation of a Single-Exam Risk Prediction Model of Severe Retinopathy of Prematurity on Premature Infants from India. *American Association for Pediatric Ophthalmology and Strabismus Annual Meeting*. March 2022. Phoenix, AZ. Oral presentation.
8. **Coyner AS\***, Campbell JP\*. Artificial Intelligence: Demystification and Applications — Getting Started in AI Research. *American Academy of Ophthalmology*. November 2021. New Orleans, LA. \*Authors contributed equally. Oral presentation.
9. **Coyner AS**, Kalpathy-Cramer J, Schelonka RL, *et al.* Prediction of TR-ROP Using a Deep Learning-derived Vascular Severity Score. *National Library of Medicine Informatics Training Conference*. June 2021. Virtual. Oral presentation.
10. **Coyner AS**, Kalpathy-Cramer J, Chen JS, *et al.* A Risk Model for Early Detection of Treatment-requiring Retinopathy of Prematurity Using a Deep Learning-derived Vascular Severity Score. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2021. Virtual. Oral presentation.
11. **Coyner AS**, Chen JS, Chan RVP, *et al.* Detectability of Synthetic Retinopathy of Prematurity Retinal Fundus Images. *American Association for Pediatric Ophthalmology and Strabismus Annual Meeting*. April 2021. Virtual. Oral presentation.
12. **Coyner AS**, Chen JS, Campbell JP, *et al.* Diagnosability of Synthetic Retinal Fundus Images for Plus Disease Detection in Retinopathy of Prematurity. *American Medical Informatics Association Annual Symposium*. November 2020. Virtual. Oral presentation.
13. **Coyner AS**, Campbell JP, Kalpathy-Cramer J, *et al.* Artificial Retinal Fundus Image Generation in Retinopathy of Prematurity. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2020. Virtual. Poster presentation.
14. **Coyner AS**, Swan R, Campbell JP, *et al.* Automated Fundus Image Quality Assessment in Retinopathy of Prematurity. *National Library of Medicine Informatics Training Conference*. May 2019. Indianapolis, IN. Oral presentation.
15. **Coyner AS**, Campbell JP, Ostmo S, *et al.* Machine Learning for Prediction of Retinopathy of Prematurity Fundus Image Quality from Clinical Data. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2019. Vancouver, BC. Poster presentation.
16. **Coyner AS**, Swan R, Brown JM, *et al.* Deep Learning for Image Quality Assessment of Fundus Images in Retinopathy of Prematurity. *American Medical Informatics Association Annual Symposium*. November 2018. San Francisco, CA. Oral presentation.
17. **Coyner AS**, Swan R, Brown JM, *et al.* Deep Learning for Image Quality Assessment of Fundus Images in Retinopathy of Prematurity. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2018. Honolulu, HI. Poster presentation.
18. **Coyner AS**, Swan R, Kalpathy-Cramer J, *et al.* Automated Image Quality Assessment for Fundus Images in Retinopathy of Prematurity. *American Medical Informatics Association Annual Symposium*. November 2017. Washington, DC. Poster presentation.
19. **Coyner AS**, Swan R, Kalpathy-Cramer J, *et al.* Automated Image Quality Assessment for Fundus Images in Retinopathy of Prematurity. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2017. Baltimore, MD. Poster presentation.
20. **Coyner AS**, Ryals RC, Ku CA, *et al.* Flibanserin, an FDA approved dual serotonin receptor agonist-antagonist, provides retinal neuroprotection from light induced damage. *Association for Research in Vision and Ophthalmology Annual Meeting*. May 2016. Seattle, WA. Poster presentation.